

SUMMARY CHARTS FOR OSEM RUN

July 10, 2026

1 SUMMARY CHARTS FOR OSEM RUN

OSEM model projects and actively trades a portfolio of corporate bonds and equities in light of commitments. The basic unit in OSEM is a cash flow. The purpose of this script is to show a how different types of cash flows are treated in the latest run.

2 Table of Contents

1. Loading the summary csv

Equity

Corporate bonds

Cash

Liability profile

2. The initial portfolio

Periodic cash flows

Terminal cash flows

Cash movement

End of period market value

3. Test 1 Return of portfolio

4. Test 2 no vanishing cash flows

Loading the summary csv

```
[29]: import pandas as pd
import matplotlib.pyplot as plt
import datetime as dt
```

```
[30]: out = pd.read_csv("Output/Results.csv", index_col=0, parse_dates=[0])
```

The initial portfolio

This section shows what the initial inputs of the run were. Currently OSEM can handle three types of financial instruments and one type of liability profile.

2.0.1 Equity

Shares in companies:

```
[31]: equity = pd.read_csv("Input/Equity_Portfolio.csv",index_col=0)
```

```
[32]: display(equity)
```

Asset_ID	Asset_Type	NACE	Issue_Date	Dividend_Yield	Frequency	Units	\
1125	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
2123	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
3232	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
1	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
2	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
3	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
4	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
5	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
6	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
7	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
8	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
9	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
10	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
11	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
12	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
13	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
14	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
15	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
16	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
17	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
18	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
19	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
20	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
21	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
22	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
23	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
24	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
25	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
26	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
27	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	
28	Equity_Share	A1.4.5	3/12/2021	0.03	1	100	
29	Equity_Share	B5.2.0	3/12/2021	0.05	1	100	
30	Equity_Share	B8.9.3	3/12/2019	0.04	1	100	

Asset_ID	Market_Price	Terminal	Default_Probability	Growth_Rate	\
1125	123.01	1	0.03	0.01	
2123	12.12	1	0.03	0.02	
3232	56.54	1	0.03	0.04	

1	77.00	1	0.03	0.01
2	78.00	1	0.03	0.02
3	1.23	1	0.03	0.04
4	3.12	1	0.03	0.01
5	45.12	1	0.03	0.02
6	96.00	1	0.03	0.04
7	94.00	1	0.03	0.01
8	92.00	1	0.03	0.02
9	96.00	1	0.03	0.04
10	94.00	1	0.03	0.01
11	92.00	1	0.03	0.02
12	96.00	1	0.03	0.04
13	94.00	1	0.03	0.01
14	92.00	1	0.03	0.02
15	96.00	1	0.03	0.04
16	94.00	1	0.03	0.01
17	92.00	1	0.03	0.02
18	96.00	1	0.03	0.04
19	94.00	1	0.03	0.01
20	92.00	1	0.03	0.02
21	96.00	1	0.03	0.04
22	94.00	1	0.03	0.01
23	92.00	1	0.03	0.02
24	96.00	1	0.03	0.04
25	94.00	1	0.03	0.01
26	92.00	1	0.03	0.02
27	96.00	1	0.03	0.04
28	94.00	1	0.03	0.01
29	92.00	1	0.03	0.02
30	96.65	1	0.03	0.04

Asset_ID	Spread_Base	Spread_Individual	Spread_Country	Spread_Sector	\
1125	0	0	0	0	
2123	0	0	0	0	
3232	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	
5	0	0	0	0	
6	0	0	0	0	
7	0	0	0	0	
8	0	0	0	0	
9	0	0	0	0	
10	0	0	0	0	
11	0	0	0	0	
12	0	0	0	0	

13	0	0	0	0
14	0	0	0	0
15	0	0	0	0
16	0	0	0	0
17	0	0	0	0
18	0	0	0	0
19	0	0	0	0
20	0	0	0	0
21	0	0	0	0
22	0	0	0	0
23	0	0	0	0
24	0	0	0	0
25	0	0	0	0
26	0	0	0	0
27	0	0	0	0
28	0	0	0	0
29	0	0	0	0
30	0	0	0	0

Asset_ID	Spread_Stress
1125	0
2123	0
3232	0
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
24	0

```

25          0
26          0
27          0
28          0
29          0
30          0

```

2.0.2 Corporate bonds

Corporate bonds:

```
[33]: bonds = pd.read_csv("Input/Bond_Portfolio.csv", index_col=0)
```

```
[34]: display(bonds)
```

	Asset_Type	NACE	Issue_Date	Maturity_Date	Notional_Amount	\
Asset_ID						
1234	Corporate_Bond	A1.4.5	3/12/2021	12/12/2026	100	
2889	Corporate_Bond	B5.2.0	3/12/2021	12/12/2028	100	
31	Corporate_Bond	B8.9.3	3/12/2019	3/12/2025	100	
1	Corporate_Bond	A1.4.5	30/06/2020	30/06/2030	100	
2	Corporate_Bond	B5.2.0	31/12/2020	31/12/2035	100	
3	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
4	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
5	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
6	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
7	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
8	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
9	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
10	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
11	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
12	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
13	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
14	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
15	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
16	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
17	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
18	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
19	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
20	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
21	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
22	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
23	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
24	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
25	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
26	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	
27	Corporate_Bond	B8.9.3	3/12/2021	12/12/2026	100	
28	Corporate_Bond	A1.4.5	3/12/2021	12/12/2028	100	
29	Corporate_Bond	B5.2.0	3/12/2019	3/12/2025	100	

30 Corporate_Bond B8.9.3 30/06/2020 30/06/2030 100

Asset_ID	Coupon_Rate	Spread_Country	Spread_Sector	Z_Spread	Spread_Stress	\
1234	0.03	0.00	0	0.010	0	
2889	0.05	0.00	0	0.010	0	
31	0.04	0.00	0	0.010	0	
1	0.01	0.01	0	0.005	0	
2	0.01	0.01	0	0.005	0	
3	0.03	0.00	0	0.010	0	
4	0.05	0.00	0	0.010	0	
5	0.04	0.00	0	0.010	0	
6	0.03	0.00	0	0.010	0	
7	0.05	0.00	0	0.010	0	
8	0.04	0.00	0	0.010	0	
9	0.03	0.00	0	0.010	0	
10	0.05	0.00	0	0.010	0	
11	0.04	0.00	0	0.010	0	
12	0.03	0.00	0	0.010	0	
13	0.05	0.00	0	0.010	0	
14	0.04	0.00	0	0.010	0	
15	0.03	0.00	0	0.010	0	
16	0.05	0.00	0	0.010	0	
17	0.04	0.00	0	0.010	0	
18	0.03	0.00	0	0.010	0	
19	0.05	0.00	0	0.010	0	
20	0.04	0.00	0	0.010	0	
21	0.03	0.00	0	0.010	0	
22	0.05	0.00	0	0.010	0	
23	0.04	0.00	0	0.010	0	
24	0.03	0.00	0	0.010	0	
25	0.05	0.00	0	0.010	0	
26	0.04	0.00	0	0.010	0	
27	0.03	0.00	0	0.010	0	
28	0.05	0.00	0	0.010	0	
29	0.04	0.00	0	0.010	0	
30	0.01	0.01	0	0.005	0	

Asset_ID	Frequency	Recovery_Rate	Default_Probability	Units	Market_Price
1234	1	0.4	0.03	100	94
2889	2	0.4	0.03	120	92
31	12	0.4	0.03	100	96
1	2	0.8	0.03	120	90
2	1	0.8	0.03	100	85
3	1	0.4	0.03	120	94
4	2	0.4	0.03	100	92
5	12	0.4	0.03	120	96

6	1	0.4	0.03	100	94
7	2	0.4	0.03	120	92
8	12	0.4	0.03	100	96
9	1	0.4	0.03	120	94
10	2	0.4	0.03	100	92
11	12	0.4	0.03	120	96
12	1	0.4	0.03	100	94
13	2	0.4	0.03	120	92
14	12	0.4	0.03	100	96
15	1	0.4	0.03	120	94
16	2	0.4	0.03	100	92
17	12	0.4	0.03	120	96
18	1	0.4	0.03	100	94
19	2	0.4	0.03	120	92
20	12	0.4	0.03	100	96
21	1	0.4	0.03	120	94
22	2	0.4	0.03	100	92
23	12	0.4	0.03	120	96
24	1	0.4	0.03	100	94
25	2	0.4	0.03	120	92
26	12	0.4	0.03	100	96
27	1	0.4	0.03	120	94
28	2	0.4	0.03	100	92
29	12	0.4	0.03	120	96
30	2	0.8	0.03	10	90

2.0.3 Cash

Initial cash position:

```
[35]: cash = pd.read_csv("Input/Cash_Portfolio.csv",index_col=0)
```

```
[36]: display(cash)
```

	Bank_Account
Asset_ID	
1	100000

2.0.4 Liability profile

An aggregated view of liabilities displayed as aggregated outflows at future dates:

```
[37]: liability = pd.read_csv("Input/Liability_Cashflow.csv",index_col=0)
```

```
[38]: display(liability)
```

	Liability_Size
Liability_Date	
1/9/2023	6
2/10/2023	92

2/11/2023	93
3/12/2023	2
3/1/2024	7
...	...
19/3/2049	14
19/4/2049	28
20/5/2049	53
20/6/2049	92
21/7/2049	41

[306 rows x 1 columns]

What is in the summary csv

During the OSEM run, pieces of information are saved into the “Summary” dataframe. The object captures the performance of the portfolio and size of cash flows in each step.

SUMMARY OBJECT

All columns

[39]: `out.head(10)`

```
[39]:
```

	Start cash	End cash	Start market value \
2023-04-29	NaN	1.000000e+05	NaN
2024-04-28	1.000000e+05	-8.731149e-11	605279.000000
2025-04-28	-8.731149e-11	2.910383e-11	560478.382092
2026-04-28	2.910383e-11	2.910383e-11	616684.521101
2027-04-28	2.910383e-11	0.000000e+00	568130.176541
2028-04-27	0.000000e+00	-7.275958e-12	625498.466625
2029-04-27	-7.275958e-12	1.164153e-10	675681.828214
2030-04-27	1.164153e-10	-4.365575e-11	825065.801716
2031-04-27	-4.365575e-11	-4.365575e-11	888114.828990
2032-04-26	-4.365575e-11	-2.619345e-10	962380.069428

	After growth market value	End market value	Portfolio return \
2023-04-29	NaN	6.052790e+05	NaN
2024-04-28	616059.287760	5.604784e+05	0.017810
2025-04-28	525750.760122	6.166845e+05	-0.061961
2026-04-28	520445.746771	5.681302e+05	-0.156058
2027-04-28	454517.349688	6.254985e+05	-0.199977
2028-04-27	625786.464798	6.756818e+05	0.000460
2029-04-27	580989.856655	8.250658e+05	-0.140143
2030-04-27	843126.394013	8.881148e+05	0.021890
2031-04-27	881258.560505	9.623801e+05	-0.007720
2032-04-26	985221.570420	1.036689e+06	0.023734

	Dividend cash flow	Coupon cash flow	Terminal cash flow \
--	--------------------	------------------	----------------------

2023-04-29	NaN	NaN	NaN
2024-04-28	11067.412743	67460.000000	0.0
2025-04-28	10313.127959	61373.754778	0.0
2026-04-28	12391.186252	53207.367169	0.0
2027-04-28	13857.489982	17077.561563	0.0
2028-04-27	19539.785618	18211.512004	0.0
2029-04-27	21621.449371	19663.556848	0.0
2030-04-27	31469.166794	884.924631	0.0
2031-04-27	33978.383914	595.536049	0.0
2032-04-26	38040.638596	282.763593	0.0

	Notional cash flow	...	UL gross premium cash flow	\
2023-04-29	NaN	...	NaN	
2024-04-28	0.000000	...	27029.633634	
2025-04-28	0.000000	...	19246.878242	
2026-04-28	117384.885451	...	19631.549716	
2027-04-28	128139.957159	...	11906.108233	
2028-04-27	0.000000	...	12144.065794	
2029-04-27	190404.159624	...	12386.779217	
2030-04-27	0.000000	...	12634.343552	
2031-04-27	33660.733208	...	12886.855751	
2032-04-26	0.000000	...	13144.414704	

	UL entry fee cash flow	UL admin fee cash flow	\
2023-04-29	NaN	NaN	
2024-04-28	540.592673	3793.970701	
2025-04-28	384.937565	2411.404578	
2026-04-28	392.630994	2121.045177	
2027-04-28	238.122165	1127.389801	
2028-04-27	242.881316	1181.738485	
2029-04-27	247.735584	1071.702912	
2030-04-27	252.686871	1151.556303	
2031-04-27	257.737115	1200.059310	
2032-04-26	262.888294	1286.767124	

	UL mortality cash flow	UL lapse cash flow	UL reserve	\
2023-04-29	NaN	NaN	720000.000000	
2024-04-28	-261137.952045	-0.000000	494380.637927	
2025-04-28	-0.000000	-0.000000	480199.013265	
2026-04-28	-0.000000	-154930.558819	267447.258029	
2027-04-28	-0.000000	-0.000000	224504.620571	
2028-04-27	-0.000000	-0.000000	235327.435196	
2029-04-27	-0.000000	-0.000000	213415.320395	
2030-04-27	-0.000000	-0.000000	229317.056760	
2031-04-27	-0.000000	-0.000000	238975.782726	
2032-04-26	-0.000000	-0.000000	256242.485718	

	Company account	UL policies in force	UL deaths	UL lapses
2023-04-29	0.000000	5.0	NaN	NaN
2024-04-28	4334.563373	4.0	1.0	0.0
2025-04-28	7130.905516	4.0	0.0	0.0
2026-04-28	9644.581688	2.0	0.0	2.0
2027-04-28	11010.093654	2.0	0.0	0.0
2028-04-27	12434.713455	2.0	0.0	0.0
2029-04-27	13754.151951	2.0	0.0	0.0
2030-04-27	15158.395125	2.0	0.0	0.0
2031-04-27	16616.191550	2.0	0.0	0.0
2032-04-26	18165.846968	2.0	0.0	0.0

[10 rows x 21 columns]

Periodic cash flows

The portfolio generates periodic cash flows from its investment portfolio and outflows from liabilities that the company committed to. Example of recurrent inflows are dividends and coupons.

```
[40]: time_axis = out.index[:-1]

div_axis = out["Dividend cash flow"][:-1]
coup_axis = out["Coupon cash flow"][:-1]
liab_axis = out["Liability cash flow"][:-1]
```

```
[41]: liab_axis
```

```
[41]: 2023-04-29    NaN
      2024-04-28    NaN
      2025-04-28    NaN
      2026-04-28    NaN
      2027-04-28    NaN
      2028-04-27    NaN
      2029-04-27    NaN
      2030-04-27    NaN
      2031-04-27    NaN
      2032-04-26    NaN
      2033-04-26    NaN
      2034-04-26    NaN
      2035-04-26    NaN
      2036-04-25    NaN
      2037-04-25    NaN
      2038-04-25    NaN
      2039-04-25    NaN
      2040-04-24    NaN
      2041-04-24    NaN
      2042-04-24    NaN
```

2043-04-24	NaN
2044-04-23	NaN
2045-04-23	NaN
2046-04-23	NaN
2047-04-23	NaN
2048-04-22	NaN
2049-04-22	NaN
2050-04-22	NaN
2051-04-22	NaN
2052-04-21	NaN
2053-04-21	NaN
2054-04-21	NaN
2055-04-21	NaN
2056-04-20	NaN
2057-04-20	NaN
2058-04-20	NaN
2059-04-20	NaN
2060-04-19	NaN
2061-04-19	NaN
2062-04-19	NaN
2063-04-19	NaN
2064-04-18	NaN
2065-04-18	NaN
2066-04-18	NaN
2067-04-18	NaN
2068-04-17	NaN
2069-04-17	NaN
2070-04-17	NaN
2071-04-17	NaN
2072-04-16	NaN
2073-04-16	NaN

Name: Liability cash flow, dtype: float64

```
[42]: fig, (ax1, ax2, ax3) = plt.subplots(3)

div_mask = div_axis > 0
coup_mask = coup_axis > 0
liab_mask = liab_axis < 0

if div_mask.any():
    ax1.stem(time_axis[div_mask], div_axis[div_mask], linefmt='grey',
    ↪label='Dividend cash flow')
    ax1.legend()
else:
    ax1.set_title('No dividend cash flows to plot')

if coup_mask.any():
```

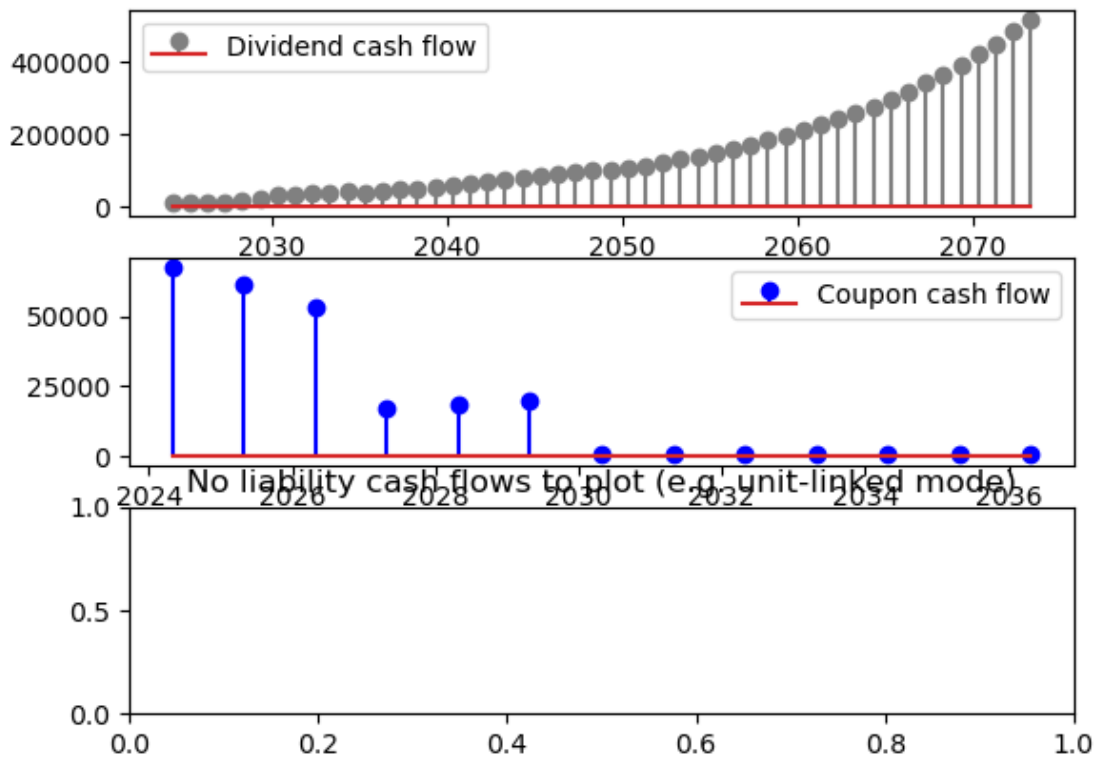
```

    ax2.stem(time_axis[coup_mask], coup_axis[coup_mask], linefmt='blue',
    ↪label='Coupon cash flow')
    ax2.legend()
else:
    ax2.set_title('No coupon cash flows to plot')

if liab_mask.any():
    ax3.stem(time_axis[liab_mask], liab_axis[liab_mask], linefmt='red',
    ↪label='Liability cash flow')
    ax3.legend()
else:
    ax3.set_title('No liability cash flows to plot (e.g. unit-linked mode)')

plt.show()

```



2.0.5 Terminal cash flows

Each asset in OSEM has a terminal value. For fixed income instruments, these are the notional payments at the maturity date, for equities, OSEM assumes that the all equity positions are sold at the end of the modelling window at market rates.

```
[43]: not_axis = out["Notional cash flow"]
      term_axis = out["Terminal cash flow"]
```

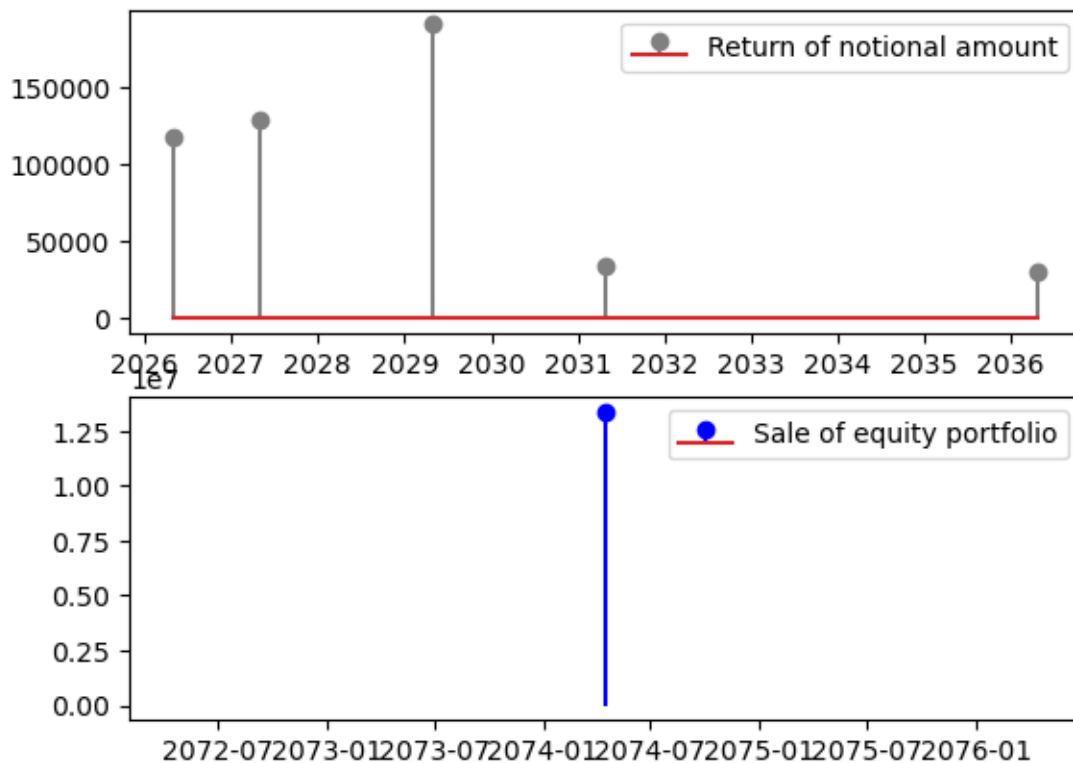
```
[44]: fig, (ax1, ax2) = plt.subplots(2)

not_mask = not_axis > 0
term_mask = term_axis > 0

if not_mask.any():
    ax1.stem(out.index[not_mask], not_axis[not_mask], linefmt='grey',
    ↪label='Return of notional amount')
    ax1.legend()
else:
    ax1.set_title('No notional cash flows to plot')

if term_mask.any():
    ax2.stem(out.index[term_mask], term_axis[term_mask], linefmt='blue',
    ↪label='Sale of equity portfolio')
    ax2.legend()
else:
    ax2.set_title('No terminal cash flows to plot')

plt.show()
```

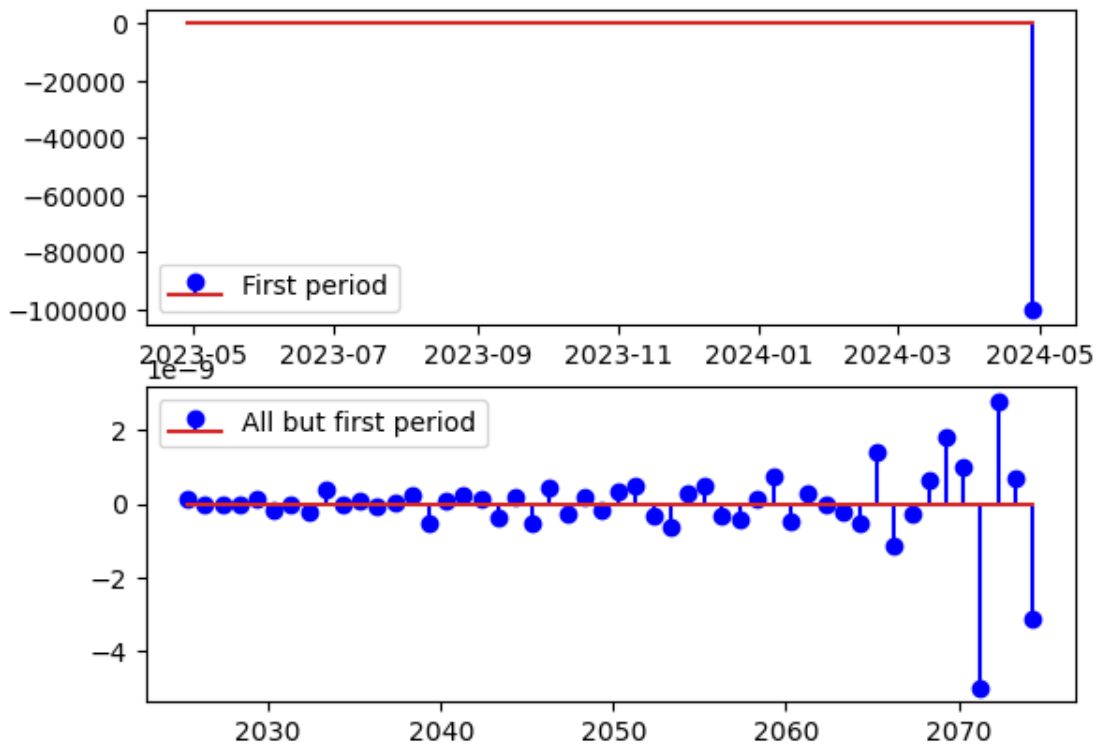


Cash movement The management of cash between periods is fundamental to OSEM. If the initial cash is high, then in the first period, the large cash position is sold in line with the asset allocation rule. This is why the initial cash is represented separately. After the first period, cash swings as the balance of inflows/outflows changes through time.

```
[45]: x_axis = out.index
      end_cash_axis = out["End cash"]
      start_cash_axis = out["Start cash"]

[46]: fig, (ax1, ax2) = plt.subplots(2)
      ax1.stem(x_axis[0:2], end_cash_axis[0:2]-start_cash_axis[0:2], linefmt='b',
      ↪label='First period')
      ax2.stem(x_axis[2:], end_cash_axis[2:]-start_cash_axis[2:], linefmt='b',
      ↪label='All but first period')

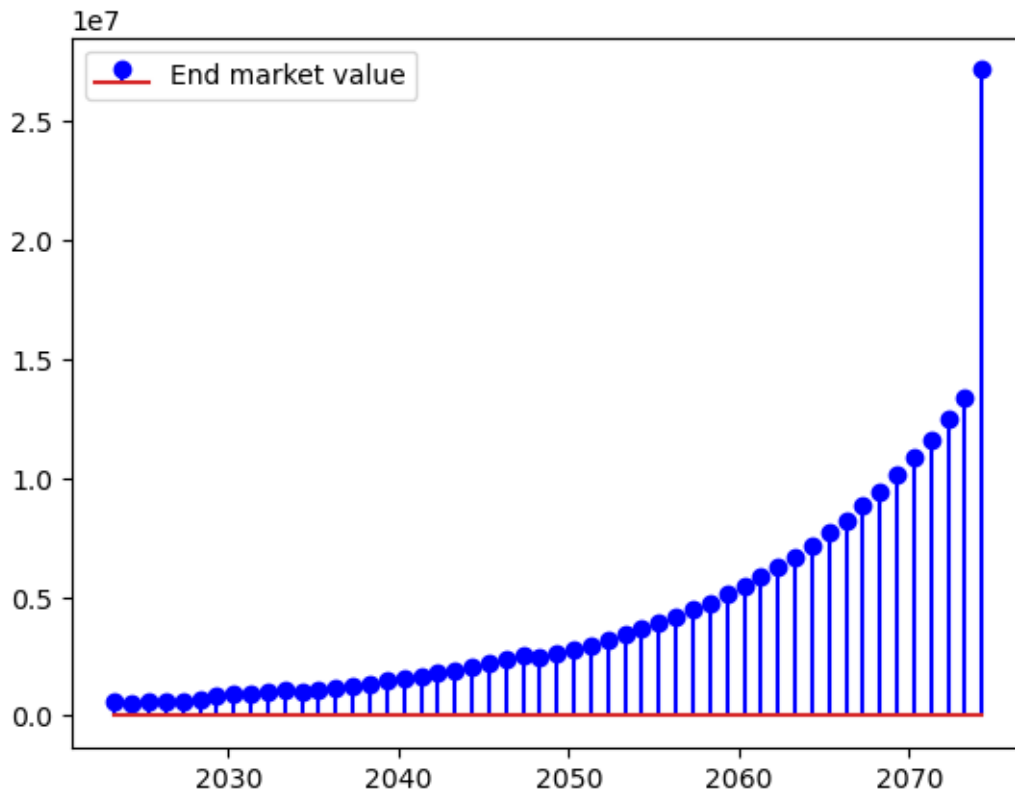
      ax1.legend()
      ax2.legend()
      plt.show()
```



End of period market value

```
[47]: start_mv_axis = out["After growth market value"]
end_mv_axis = out["End market value"]

fig, ax = plt.subplots(1)
ax.stem(out.index, end_mv_axis, linefmt='b' , label='End market value')
ax.legend()
plt.show()
```



2.1 Test 1 Return of portfolio

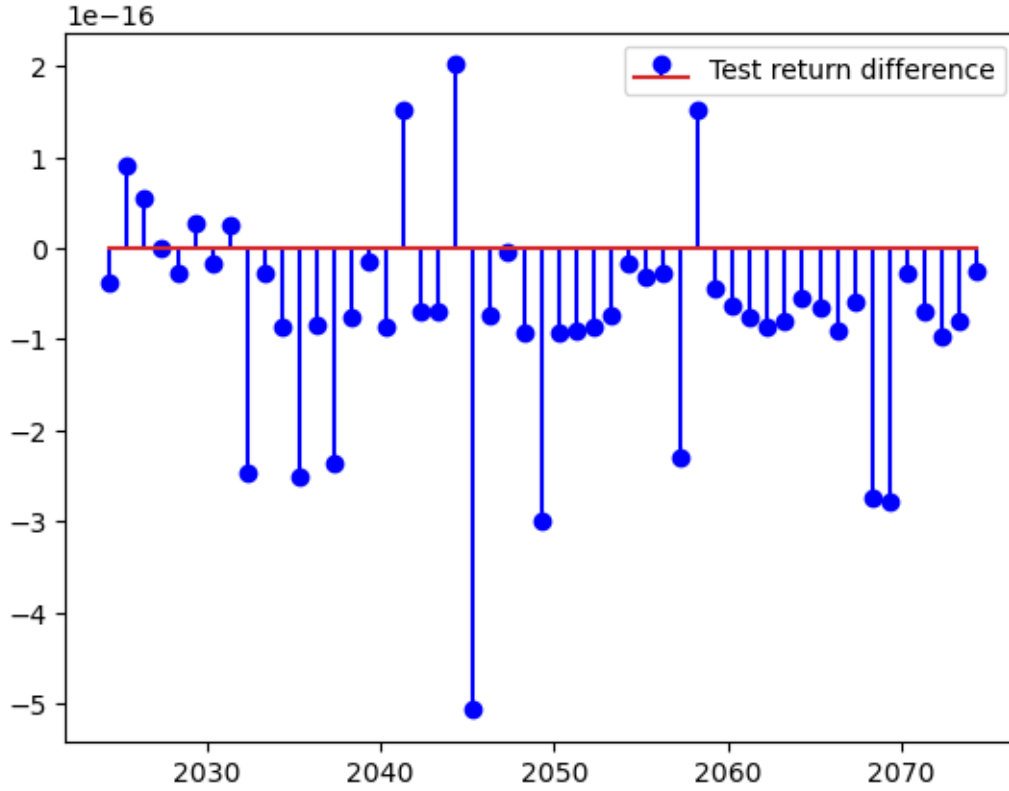
The purpose of this test is to verify if the reported return of the portfolio is equal to the increase/decrease of the total market value of the portfolio.

```
[48]: return_calculated = (out["After growth market value"]/out['Start market_
↪value']-1)[1:]

[49]: test_1 = out["Portfolio return"][1:]-return_calculated

[50]: fig, ax = plt.subplots(1)
ax.stem(out.index[1:], test_1, linefmt='b' , label='Test return difference')
ax.legend()
```

```
plt.show()
```



2.2 Test 2 no vanishing cash flows

The purpose of this test is to verify if the inflows and outflows balance at every period. This test shows that all types of cash flows are accounted correctly and there is no “leakage”.

Starting values are

- Start cash
- After growth market value

Cash flows occurring during the period and accumulating inside the bank account are * Coupon payments * Dividend payments * Notional payments of fixed income assets * Terminal value sales

The end values are: * End cash value * End market value

The test performs the following check

$$C_i^{start} + MV_i^{ag} + CF_i^a - CF_i^l = C_i^{end} + MV_i^{end}$$

Where:

$$CF_i^a := div_i + coup_i + not_i + term_i$$

And: * i is the calculation period * C_i^{start} is the initial cash position * MV_i^{ag} is the market value after portfolio growth but before any trading * CF_i^l are the accumulated outflows from company's commitments * C_i^{end} is the final cash position * MV_i^{end} is the final market value of the portfolio * div_i is the accumulated dividend inflows * $coup_i$ is the accumulated coupon inflows * not_i is the accumulated inflows from matured fixed income instruments (bonds) * $term_i$ is the accumulated inflows from equity positions sold at the end of the modelling window

Note that the asset growth occurs before trading therefore, “Cash after growth” is used as the starting value

```
[51]: cf_assets = coup_axis+div_axis+not_axis[:-1]+term_axis[:-1]

test_2 = start_cash_axis + start_mv_axis + cf_assets + liab_axis - end_mv_axis
      ↪ - end_cash_axis
```

```
[52]: fig, ax = plt.subplots(1)
ax.stem(x_axis, test_2, linefmt='b' , label='Test return difference')
ax.legend()
plt.show()
```

